R^{2}IPoints: Pursuing Rotation-Insensitive Point Representation for Aerial Object Detection

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***Abstract*— Anchor-free aerial object detection methods have recently attracted much attention due to their simplicity and efficiency. However, the performance is still unsatisfactory due to the following two main limitations. On the one hand, the anchor-free detector employs ordinary convolution layers with axis-aligned receptive fields to extract object features, resulting in lacking internal mechanisms to handle the rotation variance. On the other hand, the detector sacrifices much semantic information to achieve faster detection, leading to the inability to deal with objects' high interclass similarity and intraclass diversity. To address these issues, in this article, we present a unique anchor-free detector, termed rotation-insensitive point representation (R^{2}IPoints), of which a set of category-aware points are employed to encode the spatial and semantic information of the arbitrary-oriented objects. Specifically, we first devise a stacked rotation convolution module (SRM) to encourage the learning of R^{2}IPoints by adaptively modeling orientation-agnostic interdependencies over stochastically rotated features. Meanwhile, we further introduce a class-specific semantic enhancement module (CSM). It performs category-aware semantic activation to recalibrate features, thus enabling the point representation to be aware of object categories. Through jointly optimizing the two proposed modules in an end-to-end manner, R^{2}IPoints could simultaneously generate rotation-insensitive and category-aware point representation. Extensive experiments on the challenging DIOR and DOTA datasets demonstrate the superiority of the proposed method. We achieve 72.7% mAP on DIOR and 74.34% mAP on DOTA, surpassing the baseline method of +2.4% mAP and +2.49% mAP, respectively. The code is available at https://github.com/shnew/R2IPoints.**

***Index Terms*— Anchor-free object detection, rotation-insensitive point representation (R^{2}IPoints).**

I. Introduction

AERIAL object detection aims at predicting a bounding box with a category label from complex aerial images, which plays an important role in many interpretation applications [1]–[3]. In recent years, object detection in aerial images has achieved formidable progress [4]–[7] due to the rapid development of deep neural networks [8]–[10]. Modern object detectors can be generally grouped into anchor-based and anchor-free methods depending on whether the anchor is used.

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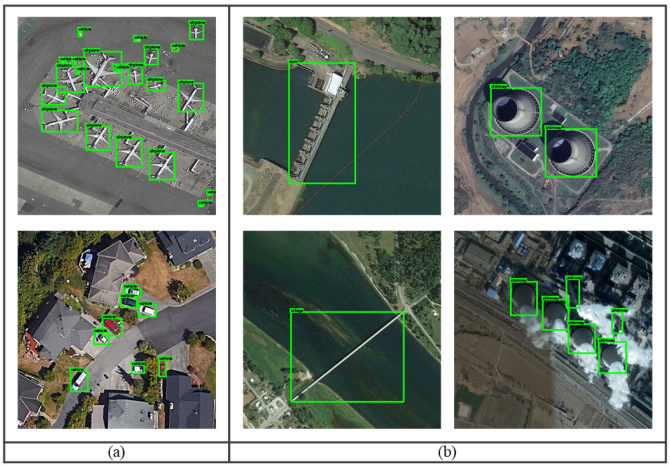


Fig. 1. Characteristics of aerial objects. (a) Arbitrary rotations. (b) High interclass similarity and intraclass diversity.

Specifically, anchor-based approaches [11]–[13], by which a set of predefined rectangular boxes are set to represent instances, have achieved remarkable results on various benchmarks. However, the detection performance is sensitive to the aspect ratios of anchors. Thus, it is necessary to predesign a set of anchors with fixed aspect ratios for different detection tasks. In addition, anchor-based methods require numerous anchors to be placed on the input images for achieving a high recall rate, which seriously increases the computational complexity.

To eliminate these problems, the anchor-free method is a natural choice, as it directly utilizes the point-based features instead of anchors to represent objects. Recently, many anchor-free detectors [14]–[21] have been designed to perform object detection in natural images and have achieved promising performance. For instance, Law and Deng [14] first predicted two heatmaps to localize the corners of bounding boxes and then adopted a specific grouping method to obtain final detection results. Inspired by it, fully convolutional one-stage object detector (FCOS) [18] adopted the center point of bounding box directly to represent objects. Similarly, representative points (RepPoints) [19] leveraged a set of representative points to compose objects and indicate important semantical features.

Despite the significant advances in natural images, anchor-free detectors suffer from large performance degradation in aerial images due to the two major challenges shown in Fig. 1. First, objects in aerial images may appear in arbitrary orientations. Unfortunately, the conventional anchor-free detectors

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employ ordinary convolution layers with axis-aligned receptive fields to extract the point feature representation of instances, thereby leading to rotation-sensitive object detectors. Consequently, inconsistent detection results may be generated for arbitrary-oriented objects of the same category. Second, aerial images usually contain objects with high interclass similarity and intraclass diversity. However, the anchor-free detector utilizes point features to represent objects, which achieves fast detection, whereas loses much semantic information. Therefore, it is impossible to capture fine-grained features with few point features to represent the diversity of object appearance, resulting in ambiguity in recognition.

In order to tackle the first problem, a stacked rotation convolution module (SRM) is elaborately designed to encourage the model to learn rotation-insensitive point representation (R^2IPoints). Specifically, we first construct a filter bank comprising several stochastic-oriented filters to capture orientation-agnostic visual patterns. It could simulate different rotation patterns of objects in the feature level. Then, we aggregate the obtained orientation-agnostic visual patterns along the channel dimension for obtaining coupled features. Finally, we model orientation-agnostic interdependencies over the coupled features to emphasize the rotation-insensitive channel features and suppress the rotation-sensitive channel features under the guidance of object supervision information. Accordingly, rotation-insensitive features can be captured no matter how the orientation of the object changes, thereby enhancing the capability of model to handle the arbitrary-oriented objects.

In order to tackle the second problem, a class-specific semantic enhancement module (CSM) is constructed as a new branch in parallel with classification and regression for further improving the category-aware capability of point representation. More specifically, it mainly performs category-aware semantic activation under binary-level semantic-agnostic supervisions. Next, the activated semantic information is embedded into the original features to enable the local class-specific regions to be aware of object categories, thus enhancing the discrimination capability against objects with high interclass similarity and intraclass diversity. Benefitted from CSM, the proposed framework not only has flexible regression like RepPoints [19] but also learns the category-aware semantic representation without affecting the feature spatial independence of classification and localization.

The cooperation of SRM and CSM formulates a unique anchor-free detector, named R^2IPoints, where SRM is designed to extract R^2IPoints and CSM aims at endowing point representation with more class-specific information. Extensive experiments on the large-scale, publicly available benchmark for object detection in optical remote sensing images (DIOR) [22] and the large-scale dataset for object detection in aerial images (DOTA) [23] demonstrate the superiority of our R2IPoints, which achieves 72.7% mAP on DIOR and 74.34% mAP on DOTA respectively. The main contributions of this article are given as follows.

1) An SRM is designed to tackle the rotation variance by emphasizing the rotation-insensitive channel features from orientation-agnostic visual patterns.
2) We introduce a CSM to perform category-aware semantic activation, which can address the ambiguity in recognition.
3) Comprehensive experimental results clearly testify that the proposed R^2IPoints outperforms the previous state-of-the-art anchor-free works.

The remainder of this article is organized as follows. Section II elaborates the related works of object detection in recent years. In Section III, the proposed method is described in detail. Section IV shows extensive experiments to demonstrate the effectiveness of the proposed method and analyzes the results. Finally, Section V draws the conclusion and indicates the future work.

II. Related Work

Object detection is a fundamental and challenging task in computer vision, which aims to predict objects with predefined categories. In the deep learning era, object detection has been extensively studied with and has achieved promising performance. Modern object detectors can be generally grouped into anchor-based methods and anchor-free methods. There are significant differences between them. Take the anchor-based detector RetinaNet [24] and the anchor-free detector FCOS [18] for example. RetinaNet presets several anchor boxes at each location to represent objects, whereas FCOS takes points at each location as the initial representation of objects. Thus, during regression, RetinaNet learns the distance between the preset anchor box and ground-truth bounding box, while FCOS directly learns the distance between the points on the feature map and the four edges of the ground-truth bounding box.

A. Anchor-Based Detectors

In early explorations, Ren *et al.* [25] introduced the concept of anchor box in faster region-based convolutional neural network (Faster-RCNN) and creatively proposed the region proposal network (RPN) to solve the problem of slow speed and low accuracy about R-CNN series [26]–[28]. After that, the anchor mechanism is widely used in numerous detectors for addressing different challenges. For instance, Cheng *et al.* [29] introduced a rotation-invariant regularizer and a Fisher discrimination regularizer on CNN features to deal with the challenges of object rotation, within-class diversity, and between-class similarity in aerial images. Gong *et al.* [30] proposed a context-aware convolutional neural network model, which integrates a context-region-of-interest mining layer into the CNN model to enrich the feature representation and mine the underlying context information among objects. Azimi *et al.* [31] proposed a network combining image cascade and feature pyramid to extract rich semantic features. Wang *et al.* [32] improved the detection performance of small objects by aggregating context information in multiscale and same-scale feature maps. Bochkovskiy *et al.* [33] developed an efficient and powerful object detection model by employing a large number of previous methods with appropriate innovations. Cheng *et al.* [35] proposed an end-to-end cross-scale feature fusion (CSFF) framework to obtain

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powerful and discriminative features for detecting multiscale objects. Yang *et al.* [36] devised a sampling fusion network to improve the sensitivity of small objects while exploring an attention network for highlighting the object features. Qian *et al.* [37] designed a novel object detection method for aerial images based on improving bounding box regression and multilevel features fusion. Deng *et al.* [38] proposed a unified and effective detector for simultaneously detecting multiclass objects in aerial images with large-scale variability. Huang *et al.* [39] proposed a novel nonlocal-aware pyramid and multiscale multitask refinement detector (NPMMRDet) for refining the misalignments of the aerial object detection tasks. Huang *et al.* [40] designed a channel separation-aggregation (CSA) structure and a dynamic receptive field (DRF) mechanism to optimize efficiency while maintaining high accuracy and introduces a diagonal support constraint head (DSC-Head) for detecting oriented bounding boxes annotated objects. Yang *et al.* [6] applied an instance-level denoising approach on the feature maps to restraint complex background interference so that the model could detect densely arranged objects.

Although the aforementioned anchor-based detectors promote the development of object detection, there are still some problems. As we all know, anchor boxes have many hyperparameters, which usually need to be carefully designed according to some prior information for achieving high performance. Meanwhile, these hyperparameters are specific to tasks so that we need to redesign these when changing tasks.

B. Anchor-Free Detectors

Accordingly, the anchor-free method has attracted much attention and achieved promising performance in natural images in recent years. Compared with anchor-based methods, anchor-free detectors have more flexible representations. For example, Law and Deng [14] formulated an object as a pair of corner points. In addition, it introduces an extra embedding feature vector to match two corner points. Then, Duan *et al.* [15] predicted extra center points on CornerNet to filter out wrong detection boxes, which largely improves the performance. In order to further improve efficiency without sacrificing accuracy, Law *et al.* [16] introduced CornerNet-Lite by combining two efficient variants of CornerNet. Ge *et al.* [21] constructed a novel anchor-free detector YOLOX-based YOLO by combining previous advanced detection methods, thus achieving a better balance between accuracy and speed. Besides, Zhou *et al.* [17] detected four extreme points and one center point of objects by using a point estimation network. FCOS [18] directly predicts the distance between the point on feature map and the boundary of the bounding box. Similarly, RepPoints [19] generates sets of representative points to formulate objects via employing deformable convolutional networks. Inspired by the above methods, some researchers have implemented object detection for aerial images in such an anchor-free fashion. Sun *et al.* [41] proposed a shape robust anchor-free network for garbage dumps detection in aerial imagery, which encourages the learning of objects with inconspicuous appearance by employing context attention. Shi *et al.* [42] devised a new multiscale centerness descriptor (MSCD) for highlighting the center of objects and a centerness detection head (CDH) for detecting objects of interest. Wang *et al.* [43] designed a dynamic gradient adjustment network (DGANet) based on CenterNet [44], which embeds dynamic gradient adjustment into the classification loss function to harmonize the quantity imbalance between easy and hard examples. Li *et al.* [45] designed a dense path aggregation feature pyramid network (DPAFPN) to enhance the feature information propagation and reuse.

Obviously, compared with the anchor-based method, the anchor-free method has less exploration in the field of complex aerial and has much room for improvement in performance. In this article, we adopt a novel anchor-free detector RepPoints [19] as our baseline method. Compared with it, we encourage networks to learn rotation-insensitive features by introducing SRM for coping with the challenge of object rotation. Moreover, class-specific semantic maps are generated by employing CSM to enhance the point feature representation, thereby addressing the problem of high interclass similarity and intraclass diversity in objects.

III. Proposed Method

The proposed R^{2}IPoints detector is shown in Fig. 2. Its central goal is to achieve a consistent and robust detection by learning abundant rotation-insensitive and category-aware point representation. Our method is built upon the classical anchor-free RepPoints detector for its simplicity and efficiency. Next, we first give a brief overview of RepPoints. This is followed by describing the proposed SRM and CSM.

A. Overview of RepPoints

The anchor-free RepPoints detector formulates an object as a set of representative points $\mathcal{R}$

$$\mathcal{R} = \{p_i = (x_i, y_i)\}_{i=1}^{n} \tag{1}$$

where p_i denotes the ith representative point and n is the number of representative points.

It consists of two nonshared localization and classification head networks. The localization head network utilizes two stages to progressively refine the object localization. In the first localization stage, the feature map $X \in \mathbb{R}^{C\times H\times W}$ is fed into common convolution layers to obtain the first set of learnable offsets $\Delta_1 = \{\Delta_1 x_i, \Delta_1 y_i\}_{i=1}^{n}$. Thus, the initial representative points can be formulated as

$$\mathcal{R}_1 = \left\{p_i^1 = \left(x_i^1, y_i^1\right)\right\}_{i=1}^{n} = \{(x + \Delta_1 x_i, y + \Delta_1 y_i)\}_{i=1}^{n} \tag{2}$$

where x and y indicate the location on a feature map. In the second localization stage, the combination of feature maps and the first set of learnable offsets is fed into a deformable convolution layer to generate the second set of learnable offsets Δ_2. Therefore, the more refined representative points can be expressed as follows:

$$\mathcal{R}_2 = \mathcal{R}_1 + \Delta_2 = \left\{\left(x_i^1 + \Delta_2 x_i, y_i^1 + \Delta_2 y_i\right)\right\}_{i=1}^{n}. \tag{3}$$

To facilitate the supervised learning, the RepPoints applies a convert function to map representative points set $\mathcal{R}$ into a pseudo box $\mathcal{B}_{\mathcal{P}}$. Here, we use a simple converting function.

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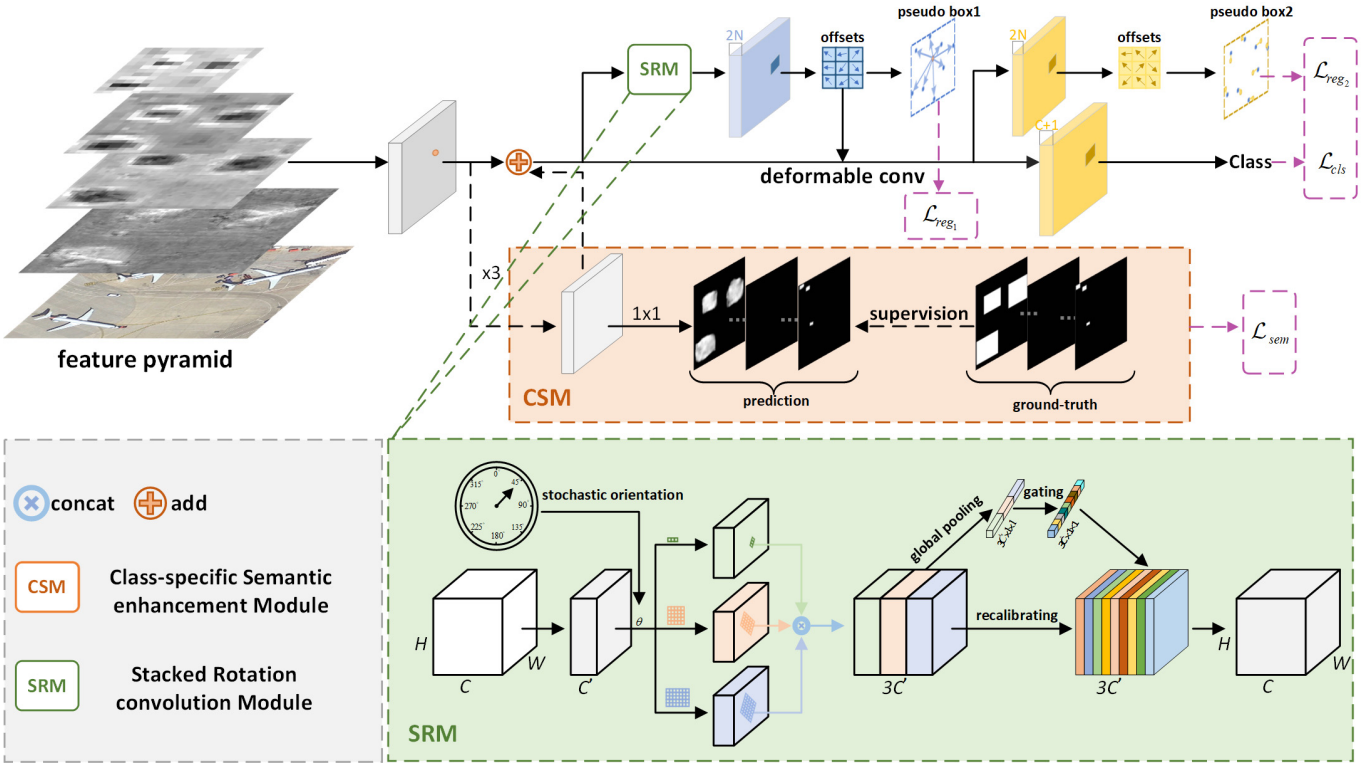


Fig. 2. Illustration of the R^{2}IPoints detector. It consists of an SRM and a CSM. Note that a backbone network and an FPN [34] are employed for feature extraction.

First, the mean and standard deviation of representative points are utilized to calculate the center point and the scale of the rectangular box. Then, two learnable parameters are used to adjust the scale of rectangular box.

In RepPoints, the object localization loss $\mathcal{L}_{\text{reg}}$ and object classification loss $\mathcal{L}_{\text{cls}}$ are exploited to guide the learning of representative points. For the classification head, it employs the focal loss [24] for classification training. For the localization head, it adopts $smoothl_1$ distance to calculate the top-left points and the bottom-right points difference between the ground-truth box $\mathcal{B}^*$ and the pseudo box $\mathcal{B}_{\mathcal{P}}$, respectively.

B. Stacked Rotation Convolution Module

The ordinary 2-D convolution operation samples feature on the input feature map X by using an axis-aligned regular paradigm. Therefore, the ordinary convolution does not achieve rotation invariance, resulting in the lack of internal mechanisms in the detector to deal with object rotation. This problem is especially evident in aerial images with plenty of oriented objects. To cope with this challenge, we elaborately construct an SRM, which consists of two parts: a filter bank comprising several stochastic-oriented filters to capture orientation-agnostic visual patterns and a channel attention block to emphasize the rotation-insensitive channel features in orientation-agnostic visual patterns.

First, we explain how rotation convolution operation works. It extracts features over the input feature map X using a rotation grid R and then sums sampled values weighted by w, where R represents the receptive field and dilation. Taking 3×3 kernel with dilation 1 for instance

$$R = \{(-1,-1),(-1,0),\ldots,(0,1),(1,1)\}. \tag{4}$$

For the rotation orientation θ, the rotation matrix $M_r(\theta)$ can be obtained according to the following formula:

$$M_r(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}. \tag{5}$$

Therefore, given the original offset $p_n \in R$ and rotation orientation θ, the additional offset is

$$\delta(\theta) = M_r(\theta) \cdot p_n - p_n. \tag{6}$$

For each location p_0 on the output feature map Y

$$Y(p_0) = \sum_{p_n \in R} w(p_n) \cdot X(p_0 + p_n + \delta(\theta)) \tag{7}$$

where p_n enumerates the locations in R and $\delta(\theta)$ denotes the offset calculated from the rotation orientation θ. Therefore, the sampling locations are depending on orientation information. As the new sampling location is generally a decimal, the output Y is calculated via bilinear interpolation.

In order to extract orientation-agnostic visual patterns, the orientation information θ in the filter bank is stochastically set between 0 and 2π. Specifically, given the input feature map $X \in \mathbb{R}^{C\times H\times W}$, a standard convolutional filter with convolution kernel 1×1 is employed to transform it into an intermediate feature map $X_m \in \mathbb{R}^{C'\times H\times W}$. Then, the feature map $X_m \in \mathbb{R}^{C'\times H\times W}$ is fed into a filter bank consisting of multiple filters with orientation information θ and specific

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receptive fields to generate the orientation-agnostic visual patterns $X_i \in \mathbb{R}^{C' \times H \times W}$, where $i \in \{1, 2, 3\}$.

We expect to highlight rotation-insensitive features and suppress rotation-sensitive features through a learnable manner. Consequently, inspired by the existing work [46], we further utilize an attention block to couple patterns in a channelwise manner. To be specific, given the orientation-agnostic visual patterns $X_i \in \mathbb{R}^{C' \times H \times W}$, a simple information aggregation $X_c \in \mathbb{R}^{(3 \times C') \times H \times W}$ is obtained by concatenating X_i in the channel direction. Therefore, rotation-sensitive features are naturally outstanding by comparison in simple information aggregation. Then, we use a global average pooling (GAP) layer, followed by two fully connected layers to obtain the channelwise map $w_c \in \mathbb{R}^{(3 \times C') \times 1 \times 1}$, which is formulated as

$$w_c = \sigma_2(W_2 \sigma_1(W_1 \cdot \mathrm{GAP}(X_c)) \tag{8}$$

where σ denotes the activate function and W refers to the parameter matrix. Hence, a rotation-insensitive feature representation $X_{\text{out}} \in \mathbb{R}^{(3 \times C') \times H \times W}$ is obtained by integrating the features of multiple branches through dot product operation as

$$X_{\text{out}} = w_c \odot X_c. \tag{9}$$

In the end, in order to facilitate the use of subsequent network, a 3×3 convolution block is exploited to restore the channel number of X_{out} to the original number C.

Therefore, the object detector can learn the rotation-insensitive representation by adaptively suppressing the rotation-sensitive channel features and emphasizing the rotation-insensitive channel features, thereby driving the network to achieve consistent prediction for arbitrary-oriented objects.

C. Class-Specific Semantic Enhancement Module

As mentioned above, anchor-free detectors sacrifice much semantic information while achieving faster detection. What is worse, in the RepPoints, the representative points tend to express the boundary characteristics of objects under the guidance of localization losses. Thus, the detector lacks sufficient fine-grained features for classification, especially when objects have high interclass similarity and intraclass diversity. The designed CSM attempts to generate discriminative and category-aware semantic information by adding an auxiliary class-specific semantic branch in parallel with the classification head and localization head.

Specifically, let $X \in \mathbb{R}^{C \times H \times W}$ denote a feature map obtained by feeding input images into the backbone network. As with classifying and locating branches, we also use three 3×3 convolution layers to obtain a class-specific feature map $\psi \in \mathbb{R}^{C \times H \times W}$. Then, the category-aware foreground score map $S_{fg} \in \mathbb{R}^{N \times H \times W}$ is obtained by employing single convolution layer with 1×1 filter, where N represents the number of target classes. Note that the supervisory information for this branch can be directly obtained from the object bounding box annotations. We exploit the weighted focal loss [47], which is defined for all object categories to guide network learning more powerful feature maps, which is denoted as

$$\mathcal{L}_{\text{sem}} = \sum_{n=1}^{N} \sum_{i=1}^{H} \sum_{j=1}^{W} \times \begin{cases} -\dfrac{w_{\text{nij}}}{M_w} \alpha \left(1 - p_{\text{nij}}\right)^{\gamma} \log\left(p_{\text{nij}}\right), & \text{if } y_{\text{nij}} = 1 \\ -\dfrac{1}{M}(1 - \alpha)\left(p_{\text{nij}}\right)^{\gamma} \log\left(1 - p_{\text{nij}}\right), & \text{otherwise} \end{cases} \tag{10}$$

where y_{nij} denotes the ground-truth foreground mask map, p_{nij} denotes the predicted nth category score at position (i, j), w_{nij} denotes the reciprocal of the object area, M_w is sum of all w_{nij}, and M is the number of the positive objects.

Finally, we aggregate class-specific semantic information into the original features, thereby endowing point features with the ability to perceive categories.

Accordingly, by exploiting CSM, the model can obtain richer class-specific semantic information to enhance the discrimination capability against objects with high interclass similarity and intraclass diversity.

Collaborating abovementioned losses constitutes the final loss of our framework, which is denoted as

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_1 \mathcal{L}_{\text{reg}_1} + \mathcal{L}_{\text{reg}_2} + \lambda_2 \mathcal{L}_{\text{sem}} \tag{11}$$

where $\mathcal{L}_{\text{reg}_1}$ and $\mathcal{L}_{\text{reg}_2}$ denote localization losses in the two stages.

IV. Experiments

A. Datasets and Evaluation Metric

A comprehensive evaluation of the proposed rotation-insensitive detector is constructed on the challenging DIOR and DOTA datasets.

The DIOR dataset contains 23 463 categories and 192 472 object instances. It is randomly divided into a train set with 11 725 images and a test set with 11 738 images. Each image is of the size 800×800 and almost every image contains more than one instance with arbitrary orientations. Meanwhile, object instances in the DIOR dataset have high interclass similarity and intraclass diversity.

The DOTA dataset includes 2806 aerial object categories and 18 828 instances. It is randomly divided into the training set with 1411 images, the validation set with 458 images, and the test set with 937 images. Each image is of the size about 800×800 to 4000×4000 pixels and objects in images exhibit a wide variety of scales, orientations, and poses. We use the same image cutting method as in [48]. Specifically, we crop a series of 1024×1024 patches from the original training and validation images with a stride of 824 but the test set with a stride of 512. After all these processes, we get 41 461 patches in total. The training patches and validation patches (i.e., 21 046 images) are applied for training and the test patches (i.e., 20 415 images) are applied for testing. Note that we perform the horizontal bounding boxes (HBBs) task on the DOTA dataset.

For the DIOR and DOTA datasets, we employ mean average precision (mAP), that is, the average value of average precision (AP) for each object category, to quantitatively

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evaluate detection performance. The AP refers to the area under the precision–recall (P–R) curve. In the P–R curve, P denotes precision and R denotes recall, which are formulated as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

where TP, FP, and FN represent the numbers of true positives, false positives, and false negatives, respectively. We use the same calculation criterion as PASCAL VOC, that is, the intersection over union (IOU) threshold with 50% is treated as the criterion.

B. Implementation Details

In our experiments, we use RepPoints as our baseline method. It employs the ResNet-101 [49], pretrained on ImageNet [50], and FPN [34] as the feature extractor. Following [19], FPN generates five feature maps. For each feature map obtained from FPN, detector predicts one bounding box with score at per location. In the experiments, n is set to 9 by default.

During training, the input resolutions of the DIOR and DOTA datasets are 800×800 and 1024×1024, respectively. Random horizontal image flipping is adopted in training. The initial learning rate is set as 0.01 for the DIOR dataset and 0.005 for the DOTA dataset. A linear warmup strategy is adopted in the first 500 iterations. For both datasets, we train our model for 24 epochs and decay the learning rate by 10 at the 20th and 22th epochs. The whole network is optimized by the stochastic gradient descent (SGD) algorithm with 0.9 momentum and 0.0001 weight decay. In the SRM, the receptive fields are 3×1, 5×5, and 7×5 to accommodate objects of divers shapes. In the CSM, the hyperparameter α is set as 0.25 and γ is set as 2. In the final loss, the weight λ_1 is set as 0.5 and λ_2 is set as 0.5.

During inference, the same input resolutions are utilized as in the training phase. After getting the preliminary bounding boxes, we use a preset score of 0.05 to filter out poor predictions. Subsequently, the IOU threshold of 0.5 is applied for non-maximum suppression (NMS) to remove duplicate boxes.

All experiments are implemented with Pytorch on ubuntu16.04, RTX 2080Ti, cuDNN v7, and CUDA 10.0.

C. Ablation Experiments

We conduct a series of ablation experiments on the DIOR dataset to demonstrate the effectiveness of the key components, including SRM and CSM in our method.

1) Effectiveness of SRM: Several experiments are designed to study the impact of the parameters in the SRM, and the results are shown in Table I. First, the effect of kernel number on performance is studied experimentally. It can be observed that when the number of convolution kernels increases from 2 to 3, the detection performance is improved by 0.5% (71.6% versus 71.1%) mAP. However, the detection performance is decreased when we increase the number of convolution kernels to 4. One possible reason is that the newly added kernel introduces numerous background information, which interferes with network learning. Accordingly, in subsequent experiments, we set the number of convolution kernels as 3. Subsequently, we set experiments to study the impact of kernel size. Results reveal that the detector achieves the best performance when the shape of kernels is 3×1, 5×5, and 7×5.

TABLE I
ABLATION STUDIES ABOUT THE PARAMETERS OF SRM ON THE DIOR DATASET

Method	kernels	mAP(%)
Baseline		70.3
SRM	3×3 , 5×5	71.1
	3×3 , 5×5 , 7×7	**71.6**
	3×3 , 5×5 , 7×7 , 9×9	71.2
	3×3 , 3×5 , 5×7	71.7
	3×1, 5×5 , 7×5	**72.0**
	3×1, 5×3 , 7×7	71.8

TABLE II
ABLATION EXPERIMENTS RESULTS ABOUT EACH COMPONENT ON THE DIOR DATASET

Baseline	√	√	√	√	√	√
OCM		√				
RCM			√			
SRM				√		√
CSM					√	√
mAP(%)	70.3	70.7	71.5	72.0	71.7	72.7

In addition, we construct three comparative experiments to further demonstrate the effectiveness of our designed SRM. The ablation results are shown in Table II. First, we construct an ordinary convolution module (OCM) that consists of three ordinary convolution layers with the same receptive fields as SRM. It gains little improvement of 0.4% (70.7% versus as 70.3%) mAP. This is because simply increasing the number of convolution layers only expands the receptive fields to some extent compared with baseline method, but the extracted features are still axis-aligned, which cannot alleviate the difficulty of detecting arbitrary-oriented objects. Then, we construct a rotation convolution module (RCM) that utilizes the rotation convolution layers instead of the ordinary convolution layers in OCM. Meanwhile, the orientation information in RCM is stochastically set. It achieves an improvement of 1.2% (71.5% versus 70.3%) mAP over the baseline method, indicating that RCM could guide the network to perform rotation-insensitive detection by simulating different rotation patterns of objects in the feature level. In the end, we add an attention block into RCM to construct the SRM, which brings an improvement of 1.7% (72.0% versus 70.3%) mAP compared with the baseline method. The result clearly demonstrates that SRM can effectively alleviate the rotation challenge of objects by learning R^2IPoints.

2) Contribution of CSM: We further design an ablation experiment to investigate the effectiveness of the CSM. As presented in Table II, R^2IPoints with CSM brings an improvement of 1.4% (71.7% versus 70.3%) mAP. It is mainly because CSM

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TABLE III
OBJECT CATEGORIES IN THE DIOR DATASET

C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
Airplane	Airport	Baseball field	Basketball court	Bridge	Chimney	Dam	Expressway service area	Expressway toll station	Golf field
C11	C12	C13	C14	C15	C16	C17	C18	C19	C20
Ground track field	Harbor	Overpass	Ship	Stadium	Storage tank	Tennis court	Train station	Vehicle	Wind mill

TABLE IV
PERFORMANCE COMPARISONS (AP AND mAP) AMONG DIFFERENT METHODS ON THE DIOR DATASET

Methods	Backbone	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	mAP
Anchor-based																						
Faster RCNN [25]	VGG16	53.6	49.3	**78.8**	66.2	28.0	70.9	62.3	69.0	55.2	68.0	56.9	50.2	50.1	27.7	**73.0**	39.8	75.2	38.6	23.6	45.4	54.1
SSD [51]	VGG16	59.5	72.7	72.4	75.7	29.7	65.8	56.6	63.5	53.1	65.3	68.6	49.4	48.1	59.2	61.0	46.6	76.3	55.1	27.4	65.7	58.6
YOLOv3 [52]	DarkNet53	**72.2**	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.4	31.1	61.1	44.9	49.7	**87.4**	70.6	68.7	**87.3**	29.4	48.3	78.7	57.1
YOLOv4 [33]	CSPDARKNet53	71.3	51.2	66.5	86.9	33.2	72.7	45.4	55.9	47.4	65.6	60.6	56.0	51.8	82.5	63.8	62.0	80.3	52.5	42.4	73.1	61.1
FPN [34]	ResNet50	54.1	71.4	63.3	81.0	42.6	72.5	57.5	68.7	62.1	73.1	76.5	42.8	56.0	71.8	57.0	53.5	81.2	53.0	43.1	80.9	63.1
Faster RCNN with FPN [25]	ResNet101	54.0	74.5	63.3	80.7	44.8	72.5	60.0	75.6	62.3	76.0	76.8	46.4	57.2	71.8	68.3	53.8	81.1	59.5	43.1	81.2	65.1
Mask RCNN with FPN [53]	ResNet101	53.9	76.6	63.2	80.9	40.2	72.5	60.4	76.3	62.5	76.0	75.9	46.5	57.4	71.8	68.3	53.7	81.0	**62.3**	43.0	81.0	65.2
RetinaNet [24]	ResNet101	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5	66.1
PANet [54]	ResNet101	60.2	72.0	70.6	80.5	43.6	72.3	61.4	72.1	66.7	72.0	73.4	45.3	56.9	71.7	70.4	62.0	80.9	57.0	47.2	84.5	66.1
CSFF [35]	ResNet101	57.2	79.6	70.1	**87.4**	46.1	76.6	62.7	82.6	73.2	78.2	**81.6**	50.7	59.5	73.3	63.4	58.5	85.9	61.9	42.9	86.9	68.0
SCRDet++ (FPN-based) [6]	ResNet50	66.3	**83.3**	74.3	87.3	**52.4**	**77.9**	**70.0**	**84.2**	**77.9**	**80.7**	81.2	**56.7**	**63.7**	73.2	71.9	**71.2**	83.4	62.2	**55.6**	**90.0**	**73.2**
Anchor-free																						
CenterNet [15]	Hourglass104	64.0	66.3	65.7	86.3	34.8	73.1	41.1	60.8	54.2	73.0	66.0	45.3	53.3	**81.3**	53.5	63.7	80.9	44.1	46.7	78.8	61.6
CornerNet [14]	Hourglass104	58.8	84.2	72.0	80.8	46.4	75.3	64.3	81.6	**76.3**	79.5	79.5	26.1	60.6	37.6	**70.7**	45.2	**84.0**	57.1	43.0	75.9	64.9
FCOS [18]	ResNet101	62.1	82.9	71.1	82.7	48.0	76.7	65.2	81.3	69.8	80.5	75.5	54.6	60.1	71.6	69.7	61.4	81.2	60.2	48.9	86.7	69.5
YOLOX [21]	CSPDARKNet53	**72.5**	80.3	78.5	84.2	49.6	77.3	54.0	78.5	64.9	74.5	75.2	**59.7**	57.5	73.1	67.2	86.3	60.3	84.4	50.4	88.7	70.8
Baseline [19]	ResNet101	59.4	85.0	75.6	84.6	48.3	73.6	68.1	82.1	68.4	79.9	79.3	55.2	60.7	71.9	68.7	61.1	81.4	65.2	50.9	86.6	70.3
Baseline [19]	Swin-tiny	62.2	85.1	76.1	84.8	48.2	77.4	72.5	84.3	70.6	80.2	80.7	55.4	62.1	72.7	68.7	**87.0**	67.9	85.3	49.4	71.8	72.1
Ours	ResNet101	67.9	85.9	76.2	85.4	51.5	78.0	73.1	85.1	72.5	81.0	77.3	57.3	62.4	71.8	69.7	67.7	81.4	68.4	**52.2**	**89.0**	72.7
Ours	Swin-tiny	69.2	**88.4**	**78.9**	**86.4**	**51.6**	**80.0**	**76.3**	**86.8**	72.4	**84.3**	**81.6**	56.3	**63.3**	77.9	70.1	86.9	72.9	**86.6**	51.4	71.8	**74.6**

embeds the activated category-aware semantic information into original feature maps to enable the point representation to be aware of category information. Therefore, with the help of CSM, the network greatly enhances the ability to distinguish the high interclass similarity and intraclass diversity.

Collaborating SRM with CSM can bring a significant improvement of 2.4% (72.7% versus 70.3%) mAP, which clearly indicates the effectiveness of R^{2}IPoints.

D. Comparison With Previous Methods

We evaluate the proposed R^{2}IPoints detector on the DIOR and DOTA datasets to provide comprehensive comparisons with previous methods in terms of mAP.

Table III imports the object categories of DIOR dataset. Quantitative comparisons with both anchor-free and anchor-based methods on the large DIOR dataset are shown in Table IV. Without bells and whistles, our method achieves 72.7% mAP, obtaining competitive detection performance compared with anchor-based methods and surpassing all existing anchor-free methods. Compared with the baseline method, our R^{2}IPoints achieves significant performance improvement on almost all class, especially for class "Airplane" (+8.5%), "Bridge" (+3.2%), "Chimney" (+4.4%), "Dam" (5.0%), "Expressway toll station" (+4.1%), "Storage tank" (6.6%), and "Train station" (3.2%). These excellent improvements can be attributed to the following aspects.

1) SRM effectively extracts rotation-insensitive point features by capturing interdependencies over diverse orientation-agnostic visual patterns. Thus, objects with arbitrary orientations (i.e., airplane, baseball diamond, and vehicle) can be easily distinguished.
2) CSM enables the point representation to be aware of object categories by introducing semantic information into original features. Accordingly, R^{2}IPoints can correctly identify objects with high interclass similarity (i.e., bridge, dam, overpass, and train station) and intraclass diversity (i.e., chimney, expressway toll station, and storage tank). In addition, using the tiny version of Swin-Transformer [55] (Swin-tiny) as a backbone, R^{2}IPoints still achieves an improvement of 2.5% (74.6% versus 72.1%) mAP.

Table V reports the full name and abbreviation object categories of DOTA dataset. Quantitative comparisons for each class with previous methods on the challenging DOTA dataset are shown in Table VI. Note that the detection results are

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TABLE V
OBJECT CATEGORIES IN THE DOTA DATASET

PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC
Plane	Baseball diamond	Bridge	Ground field track	Small vehicle	Large vehicle	Ship	Tennis court	Basketball court	Storage tank	Soccer ball field	Roundabout	Harbor	Swimming pool	Helicopter

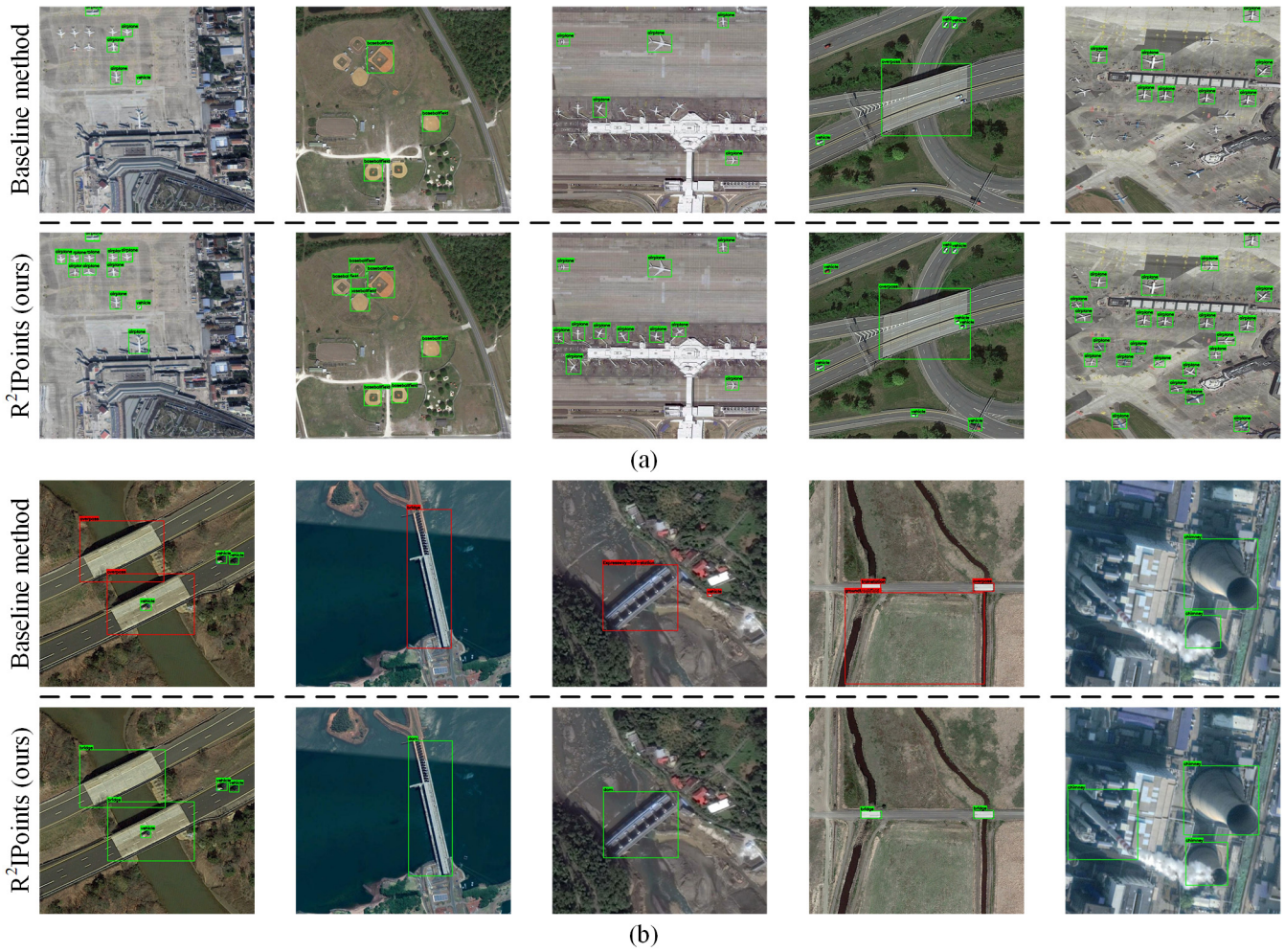


Fig. 3. Some qualitative comparison examples between the baseline method and our R^{2}IPoints. Note that we use green to indicate correct detection results and red for wrong results. (a) Instance missing due to the object arbitrary rotations. (b) Incorrect detection and instance missing due to the high interclass similarity and intraclass diversity.

calculated by the official DOTA evaluation server. As can be seen, our R^{2}IPoints achieves 74.34% mAP, which is almost on-par with anchor-based methods and outperforms all existing anchor-free approaches. Compared to the baseline method, our R^{2}IPoints obtains a large improvement of 2.49% (74.34% versus 71.85%) mAP. We also conduct experiments on DOTA datasets utilizing the tiny version of Swin-Transformer [55] (Swin-tiny) as a backbone. We can observe that the proposed R^{2}IPoints still brings 2.78% (75.85% versus 73.07%) mAP improvement compared with baseline work, which further demonstrates the effectiveness of our model. The reason of the large improvement is closely related to the proposed SRM and CSM, which facilitates the network to learn a set of rotation-insensitive and category-aware points to effectively encode the spatial and semantic information of objects.

E. Qualitative Results

We show some qualitative results comparison between the baseline method and our R^{2}IPoints in Fig. 3. As shown in Fig. 3(a), some instances in the aerial images, such as airplanes baseball fields and vehicles, appear in random directions. However, the baseline method, because it does not have rotation invariance, cannot completely detect all oriented objects belonging to the same category in an image. With the help of our designed SRM, our model could capture rotation-insensitive features and thus make consistent and correct detection results for objects in the image no matter how their orientation changes. Besides, as shown in Fig. 3(b), objects generally have high interclass similarity and intraclass diversity. For example, some object classes, such as bridge, overpass, dam, and train station, have very

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TABLE VI

PERFORMANCE COMPARISONS (AP AND mAP) AMONG DIFFERENT METHODS ON THE HBB TASK OF DOTA DATASET

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Anchor-based																	
SSD [51]	ResNet101	85.37	32.83	23.02	38.82	33.44	62.84	52.31	87.30	39.40	67.33	52.29	34.83	65.21	55.73	22.16	50.19
Faster RCNN [25]	ResNet101	80.32	77.55	32.86	68.13	53.66	52.49	50.04	90.41	75.05	59.59	57.00	49.81	61.69	56.46	41.85	60.64
RetinaNet [24]	ResNet101	72.99	68.37	**65.96**	**76.22**	25.51	62.78	68.59	84.20	62.33	51.31	57.78	**80.90**	65.96	57.81	48.50	63.27
YOLOv3 [52]	Darknet53	**93.91**	68.78	45.93	51.92	50.12	60.67	**85.56**	**93.88**	66.82	83.47	52.45	45.01	74.03	55.85	56.68	65.67
YOLOv4 [33]	CSPDARKNet53	88.8	78.1	48.3	56.0	70.7	77.4	77.8	90.6	76.8	81.1	41.4	64.9	76.2	75.1	56.3	70.7
FPN [34]	ResNet101	88.70	75.10	52.60	59.20	69.40	78.80	84.50	90.60	81.30	82.60	52.50	62.10	76.60	66.30	60.10	72.00
ICN [31]	ResNet101	89.97	77.71	53.38	73.26	73.46	65.02	78.22	90.79	79.05	84.81	57.20	62.11	73.45	70.22	58.08	72.45
IoU-Adaptive R-CNN [56]	ResNet101	88.62	80.22	53.18	66.97	**76.30**	72.59	84.07	90.66	80.95	76.24	57.12	66.65	**84.08**	66.36	56.85	72.72
SCRDet [36]	ResNet101	90.18	81.88	55.30	73.29	72.09	77.65	78.06	90.91	82.44	**86.39**	64.53	63.45	75.77	78.21	60.11	75.35
FADe [57]	ResNet101	90.15	78.60	51.92	75.23	73.60	71.27	81.41	90.85	83.94	84.77	58.91	65.65	76.92	79.36	**68.17**	75.38
SCRDet++ (FPN-based) [6]	ResNet101	90.01	**82.32**	61.94	68.62	69.62	**81.17**	78.83	90.86	**86.32**	85.10	**65.10**	61.12	77.69	**80.68**	64.25	**76.24**
Anchor-free																	
DPAFPN [45]	ResNet101	80.23	73.17	48.81	55.13	55.97	49.46	57.76	81.12	66.57	75.75	45.45	65.82	64.88	72.38	51.36	62.92
FCOS [18]	ResNet101	89.44	80.96	50.86	64.69	70.34	65.31	69.69	90.88	79.70	81.96	57.20	61.12	72.05	72.54	46.83	70.23
YOLOX [21]	CSPDARKNet53	87.56	**84.29**	53.26	68.48	70.16	69.30	86.33	90.88	**85.22**	**86.45**	51.48	62.71	76.64	77.16	39.86	72.65
CenterNet [15]	Hourglass104	**97.37**	78.56	45.39	53.39	62.16	**80.24**	**90.30**	**94.58**	66.11	85.75	**64.86**	**69.02**	**78.86**	75.63	**66.82**	73.94
Baseline [19]	ResNet101	89.33	77.24	54.00	66.17	76.55	69.46	79.19	90.89	80.89	84.29	54.07	64.11	76.58	76.65	38.35	71.85
Baseline [19]	Swin-tiny	89.45	78.86	55.64	68.74	76.39	68.58	79.18	90.89	77.78	83.12	53.90	65.08	75.32	80.39	52.83	73.07
Ours	ResNet101	89.50	81.83	**56.00**	**73.45**	77.35	68.21	79.23	90.87	79.99	84.06	61.84	68.39	76.65	75.19	52.56	74.34
Ours	Swin-tiny	89.20	83.83	55.78	72.50	**77.51**	69.98	79.28	90.90	82.07	85.95	59.94	67.24	77.04	**82.48**	64.03	**75.85**



Fig. 4. Qualitative detection results on the DIOR dataset (72.7% mAP). Green color rectangles indicate detection results.

similar appearances. Therefore, it is difficult for baseline method to generate correct category labels. Meanwhile, chimney instances have high intraclass diversity. As a result, the baseline method often fails to identify all instances in the image. Benefitting from the proposed CSM, our detector enables the point representation to be aware of object categories. Accordingly, as shown in the second line of Fig. 3(b), our R^{2}IPoints can correctly identify object categories and find out all instances. At the same time, some detection results are shown in Figs. 4 and 5 to qualitatively analyze the effectiveness of the proposed R^{2}IPoints on the DIOR and DOTA datasets, respectively. Note that different colors represent different categories. As can be found, our proposed method can correctly locate and classify objects with different

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Fig. 5. Qualitative detection results on the DOTA dataset (74.34% mAP). Different colors rectangles indicate different classes.

orientations, high interclass similarity, and intraclass diversity, which clearly illustrates the superiority of our model.

V. Conclusion

In this article, we propose a unique anchor-free detector called R^{2}IPoints with SRM and CSM. Specifically, the SRM simulates the rotation in feature level and then models orientation-agnostic interdependencies over different rotation patterns to encourage the learning of R^{2}IPoints. Furthermore, the CSM could enable point representation to be aware of object categories by introducing activated semantic information into original features. Through jointly optimizing two proposed modules in an end-to-end manner, R^{2}IPoints could simultaneously obtain rotation-insensitive and category-aware point representation to achieve a consistent and robust detection. Comprehensive experiments on the challenging DIOR and DOTA datasets demonstrate the superiority of the proposed method in both quantitative and qualitative ways.

Despite its demonstrated benefits, our R^{2}IPoints also has trouble in detecting small objects with dense arrangements. The main reason is that the commonly used IoU matching mechanism is very sensitive to position deviation of small objects. To alleviate it, the future work will propose a new matching metric for small object detection.

References

[1] S. Tuermer, F. Kurz, P. Reinartz, and U. Stilla, "Airborne vehicle detection in dense urban areas using HoG features and disparity maps," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 6, pp. 2327–2337, Dec. 2013.

[2] P. Zhong and R. Wang, "A multiple conditional random fields ensemble model for urban area detection in remote sensing optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 3978–3988, Dec. 2007.

[3] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[4] G. Cheng, R. Li, C. Lang, and J. Han, "Task-wise attention guided part complementary learning for few-shot image classification," *Sci. China Inf. Sci.*, vol. 64, no. 2, pp. 1–14, 2021.

[5] X. Feng, J. Han, X. Yao, and G. Cheng, "TCANet: Triple context-aware network for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6946–6955, Aug. 2021.

[6] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," 2017, *arXiv:1702.07054*.

[7] X. Yao, X. Feng, J. Han, G. Cheng, and L. Guo, "Automatic weakly supervised object detection from high spatial resolution remote sensing images via dynamic curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 675–685, Jan. 2021.

[8] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2012, pp. 1097–1105.

[9] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.

[10] T. Yang, S. Zhu, and C. Chen, "GradAug: A new regularization method for deep neural networks," 2020, *arXiv:2006.07989*.

[11] J. Wang, K. Chen, S. Yang, C. C. Loy, and D. Lin, "Region proposal by guided anchoring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2965–2974.

[12] Z. Ge, S. Liu, Z. Li, O. Yoshie, and J. Sun, "OTA: Optimal transport assignment for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 303–312.

[13] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

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[14] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 734–750.

[15] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[16] H. Law, Y. Teng, O. Russakovsky, and J. Deng, "CornerNet-lite: Efficient keypoint based object detection," 2019, *arXiv:1904.08900*.

[17] X. Zhou, J. Zhuo, and P. Krahenbuhl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.

[18] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[19] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9657–9666.

[20] H. Qiu *et al.*, "BorderDet: Border feature for dense object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 549–564.

[21] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding Yolo series in 2021," 2021, *arXiv:2107.08430*.

[22] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[23] G.-S. Xia *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[24] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[25] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 91–99.

[26] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[27] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Jul. 2015.

[28] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[29] G. Cheng, J. Han, P. Zhou, and D. Xu, "Learning rotation-invariant and Fisher discriminative convolutional neural networks for object detection," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 265–278, Jan. 2019.

[30] Y. Gong *et al.*, "Context-aware convolutional neural network for object detection in VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 34–44, Jan. 2019.

[31] S. M. Azimi, E. Vig, R. Bahmanyar, M. Körner, and P. Reinartz, "Towards multi-class object detection in unconstrained remote sensing imagery," in *Proc. Asian Conf. Comput. Vis.*, 2018, pp. 150–165.

[32] P. Wang, X. Sun, W. Diao, and K. Fu, "FMSSD: Feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3377–3390, Dec. 2020.

[33] A. Bochkovskiy, C.-Y. Wang, and H.-Y. Mark Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[34] T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.

[35] G. Cheng, Y. Si, H. Hong, X. Yao, and L. Guo, "Cross-scale feature fusion for object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 431–435, Mar. 2020.

[36] X. Yang *et al.*, "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8232–8241.

[37] X. Qian, S. Lin, G. Cheng, X. Yao, H. Ren, and W. Wang, "Object detection in remote sensing images based on improved bounding box regression and multi-level features fusion," *Remote Sens.*, vol. 12, no. 1, p. 143, Jan. 2020.

[38] Z. Deng, H. Sun, S. Zhou, J. Zhao, L. Lei, and H. Zou, "Multi-scale object detection in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 3–22, Nov. 2018.

[39] Z. Huang, W. Li, X.-G. Xia, X. Wu, Z. Cai, and R. Tao, "A novel nonlocal-aware pyramid and multiscale multitask refinement detector for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022.

[40] Z. Huang, W. Li, X.-G. Xia, H. Wang, F. Jie, and R. Tao, "LO-det: Lightweight oriented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[41] X. Sun, Y. Liu, Z. Yan, P. Wang, W. Diao, and K. Fu, "SRAF-Net: Shape robust anchor-free network for garbage dumps in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6154–6168, Jul. 2021.

[42] L. Shi, L. Kuang, X. Xu, B. Pan, and Z. Shi, "CANet: Centerness-aware network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[43] P. Wang, Y. Niu, R. Xiong, F. Ma, and C. Zhang, "DGANet: Dynamic gradient adjustment anchor-free object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 9, p. 1642, Apr. 2021.

[44] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[45] Y. Li, X. Pei, Q. Huang, L. Jiao, R. Shang, and N. Marturi, "Anchor-free single stage detector in remote sensing images based on multiscale dense path aggregation feature pyramid network," *IEEE Access*, vol. 8, pp. 63121–63133, 2020.

[46] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[47] Y. Chen, Z. Zhang, Y. Cao, L. Wang, S. Lin, and H. Hu, "RepPoints v2: Verification meets regression for object detection," 2020, *arXiv:2007.08508*.

[48] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[49] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[50] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[51] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2016, pp. 21–37.

[52] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[53] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," in *Proc. ICCV*, Oct. 2017, pp. 2961–2969.

[54] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[55] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[56] J. Yan, H. Wang, M. Yan, D. Wenhui, X. Sun, and H. Li, "IoU-adaptive deformable R-CNN: Make full use of IoU for multi-class object detection in remote sensing imagery," *Remote Sens.*, vol. 11, no. 3, p. 286, Feb. 2019.

[57] C. Li, C. Xu, Z. Cui, D. Wang, T. Zhang, and J. Yang, "Feature-attentioned object detection in remote sensing imagery," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 3886–3890.

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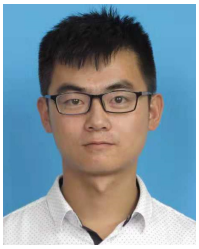
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